

FAMILY STUDIES

OVERVIEW

In the social sciences and humanities curriculum, family studies is a multidisciplinary subject area that encompasses four areas: fashion and housing; food and nutrition; general family studies; and raising and caring for children. The topics covered in these areas include, among others: textile production; the functions and design of clothing; the fashion industry; issues related to housing; interior design; the role of nutrition in health; the relationship between food and culture; human and family interactions and development; life management skills; various types of relationships; and considerations related to raising and caring for children.

In family studies courses, students learn about laws and policies that affect individuals and families in Canada and around the world. They also become familiar with the theoretical perspectives and practical research that underpin our understandings of individual and family development. Family studies courses also integrate theory and practical skills. Many expectations can be achieved through practical experiences both in the classroom and in co-op placements, some of which may be affiliated with Specialist High Skills Majors programs.

All courses in family studies encourage students to develop critical and creative thinking skills. Students are given opportunities to deepen their understanding of relevant issues and to develop practical skills, including research and inquiry skills. Students are encouraged to explore a range of perspectives and approaches and to develop the habits of mind that enhance individual, family, and community well-being and contribute to lifelong learning.

FOOD AND NUTRITION

Nutrition and Health, Grade 12

College Preparation

HFA4C

This course focuses on the relationship between nutrition and health at different stages of life and on global issues related to food production. Students will investigate the role of nutrition in health and disease and assess strategies for promoting food security and environmental responsibility. Students will learn about healthy eating, expand their repertoire of food-preparation techniques, and refine their ability to use social science research and inquiry methods to investigate topics related to nutrition and health.

Prerequisite: Any university, college, or university/college preparation course in social sciences and humanities, English, or Canadian and world studies

A. RESEARCH AND INQUIRY SKILLS

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1. Exploring:** explore topics related to nutrition and health, and formulate questions to guide their research;
- A2. Investigating:** create research plans, and locate and select information relevant to their chosen topics, using appropriate social science research and inquiry methods;
- A3. Processing Information:** assess, record, analyse, and synthesize information gathered through research and inquiry;
- A4. Communicating and Reflecting:** communicate the results of their research and inquiry clearly and effectively, and reflect on and evaluate their research, inquiry, and communication skills.

SPECIFIC EXPECTATIONS

A1. Exploring

Throughout this course, students will:

- A1.1** explore a variety of topics related to nutrition and health (*e.g., nutritional needs throughout the lifespan, nutritional status of different groups*) to identify topics for research and inquiry
- A1.2** identify key concepts (*e.g., through discussion, brainstorming, use of visual organizers*) related to their selected topics
- A1.3** formulate effective questions to guide their research and inquiry
Teacher prompt: “If you were researching the risks and benefits of drinking protein shakes, why might it be important to compare the effects on athletes and non-athletes?”

A2. Investigating

Throughout this course, students will:

- A2.1** create appropriate research plans to investigate their selected topics (*e.g., outline purpose and method; identify sources of information; develop research tools such as surveys or questionnaires*), ensuring that their plans follow guidelines for ethical research
Teacher prompts: “What steps can you take to ensure that you are not asking leading questions in your survey?” “For which purposes might websites such as Wikipedia be adequate sources

of information? What are the risks of using websites such as Wikipedia?”

- A2.2** locate and select information relevant to their investigations from a variety of primary sources (*e.g., interviews, surveys, observations, field research, data sets from Statistics Canada*) and/or secondary sources (*e.g., research reports, textbooks, advertisements, brochures, newspaper and magazine articles, websites*)

Teacher prompts: “What is the difference between primary and secondary sources in social science? How can you determine whether a source is a primary or secondary source?” “What criteria would you use to choose the best secondary source for your research question?”

- A2.3** based on preliminary research, for each investigation formulate a hypothesis, thesis statement, or research question, and use it to focus their research

A.3 Processing Information

Throughout this course, students will:

- A3.1** assess various aspects of information gathered from primary and secondary sources (*e.g., accuracy, relevance, reliability, inherent values and bias, voice*)
Teacher prompts: “What strategies can you use to determine the relevance of the information

you have gathered?" "If two information sources contradict each other, how might you determine which is more reliable?" "What is your own personal connection to the research, and how does this affect your interpretation of the information gathered?"

- A3.2** record and organize information and key ideas using a variety of formats (*e.g., journals, logs, report outlines, notes, graphic organizers, audio/visual/digital records*)
- A3.3** analyse and interpret research information (*e.g., compare information from various sources; make connections; identify gaps that necessitate further research*)
- A3.4** demonstrate academic honesty by documenting the sources of all information generated through research
- A3.5** synthesize findings and formulate conclusions (*e.g., weigh and connect information to determine the answer to their research question*)

A4. Communicating and Reflecting

Throughout this course, students will:

- A4.1** use an appropriate format (*e.g., brochure, flyer, poster, report, multimedia presentation*) to communicate the results of their research and inquiry effectively for a specific purpose and audience
 - A4.2** use terms relating to nutrition and health correctly (*e.g., macronutrient, micronutrient, nutrient deficiency, nutrient retention, food security, water potability, functional food*)
 - A4.3** clearly communicate the results of their inquiries (*e.g., write clearly, organize ideas logically, use language conventions properly*), and follow APA conventions for acknowledging sources (*e.g., generate a reference list in APA style*)
 - A4.4** demonstrate an understanding of the general research process by reflecting on and evaluating their own research, inquiry, and communication skills
- Teacher prompts:** "How might the research methods you used have affected the results you obtained?" "How did the primary and/or secondary sources you used confirm what you already knew and understood about the topic? How did your primary and/or secondary sources contradict what you thought was true?" "What steps might you take to enhance your research/inquiry skills?"

B. NUTRITION AND HEALTH

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1. Nutrients:** demonstrate an understanding of nutrients and their connection to physical health;
- B2. Food Guides:** demonstrate an understanding of Canada's Food Guide and its role in promoting physical health;
- B3. Energy Balance:** demonstrate an understanding of the physical processes involved in maintaining energy balance;
- B4. Nutritional Status:** demonstrate an understanding of their nutrient intake and of factors that affect the nutritional status of individuals and groups.

SPECIFIC EXPECTATIONS

B1. Nutrients

By the end of this course, students will:

- B1.1** describe sources and functions of macronutrients (i.e., carbohydrates, fats, proteins), micronutrients (i.e., vitamins, minerals), and water

Teacher prompt: "What are the types of fat that the body needs most? What are good sources of these fats?"

- B1.2** describe the causes and symptoms of nutrient deficiencies (e.g., rickets, pellagra, goitre, anaemia, osteoporosis, scurvy, kwashiorkor, marasmus, beriberi) and excesses (e.g., iron toxicity, fluorosis)

Teacher prompts: "Which nutrients are needed to prevent osteoporosis?" "What is a goitre and why are goitres less common in Canada now than fifty years ago?"

- B1.3** describe the nutrient content of specific foods, using available food and nutrition information (e.g., Nutrition Facts tables, food company nutrition information, nutrient-values databases, information provided by health and nutrition professionals)

Teacher prompt: "What criteria would you use to determine the best source of nutrients when comparing similar foods?"

- B1.4** plan and prepare a food item or items to ensure optimal nutrient content and retention (e.g., choose nutrient-dense foods; steam rather than boil vegetables)

B2. Food Guides

By the end of this course, students will:

- B2.1** describe the evolution of Canada's Food Guide over time (e.g., changes in name, food groups, objectives, serving amounts, key recommendations)

Teacher prompts: "What are the main differences between the current and previous versions of Canada's Food Guide?" "What new recommendations appear?"

- B2.2** outline the main nutrients found in each of the food groups in Canada's Food Guide (e.g., carbohydrates in the Grain Products group, protein in the Meat and Alternatives and Milk and Alternatives groups)

Teacher prompt: "In addition to protein, what other nutrients are found in the Meat and Alternatives group?"

- B2.3** describe research findings that support the recommendations and guidelines in Canada's Food Guide (e.g., research showing that whole-grain products have more vitamins, minerals, and fibre than comparable non-whole-grain products)

- B2.4** compare the key recommendations in Canada's Food Guide to those in food guides from other countries (e.g., Dietary Guidelines for Americans, Mediterranean Food Guide, Chinese

Food Guide) and food guides designed for special groups (e.g., vegetarians, vegans, diabetics)

Teacher prompts: “Why might particular groups need specialized food guides?” “What similarities are there among the different food guides?”

- B2.5** prepare a food item or items to reflect specific recommendations in Canada’s Food Guide (e.g., *reduce fat or sodium; use whole grains, dark green or orange vegetables, beans or lentils*)

B3. Energy Balance

By the end of this course, students will:

- B3.1** describe the processes of food digestion, absorption, and metabolism

Teacher prompt: “What are the main organs that are directly involved in food digestion?”

- B3.2** analyse foods to determine their macronutrient content (i.e., the percentage of calories from protein, fat, and carbohydrates)

- B3.3** explain the concept of energy balance, and describe how energy balance can be achieved (e.g., *by changing the quantity and types of food eaten; by changing the type, duration, or intensity of exercise*)

- B3.4** explain how various factors affect calorie expenditure (e.g., *duration and intensity of exercise, body composition, basal metabolic rate, type of activity*)

Teacher prompts: “Which types of exercise use the most calories?” “How can two people who do the same exercise for the same length of time, burn different numbers of calories?”

B4. Nutritional Status

By the end of this course, students will:

- B4.1** compare their own nutrient intake to that recommended in current Canadian guidelines for people of their age, gender, and lifestyle (e.g., *Dietary Reference Intakes*)

Teacher prompts: “How does your daily intake of vitamin E compare with the amount recommended in Canada’s Dietary Reference Intakes?” “What improvements can you make to your diet?”

- B4.2** compare their own nutrient intake with that of various population groups in Canada

Teacher prompts: “How does your nutrient intake compare to that of an average teenager living in another part of Canada?” “How does your nutrient intake compare to that of an average eighty-year-old in Canada?” “What factors should you take into account when comparing people’s nutritional status?”

- B4.3** identify factors that can contribute to the poor nutritional status of people in Canada and around the world (e.g., *genetic propensity to nutrition-related diseases such as diabetes; decaying infrastructure; natural disasters*)

Teacher prompts: “In which countries are people more likely to have poor nutritional status?” “Why do some First Nation communities in Canada have such limited access to safe drinking water? How does limited access to safe drinking water affect the nutritional status of some First Nation groups?”

- B4.4** plan and prepare a food item or items to address a specific nutritional deficiency common to Canadians (e.g., *prepare a high-fibre snack to address a lack of dietary fibre*)

C. EATING PATTERNS AND TRENDS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1. Nutrition throughout the Lifespan:** demonstrate an understanding of food- and nutrition-related issues at different stages in the lifespan;
- C2. Nutrition and Disease:** demonstrate an understanding of the relationships between nutrition, health, and disease;
- C3. Trends and Patterns in Food and Nutrition:** demonstrate an understanding of current Canadian trends and patterns in nutritional guidelines and in food production and consumption.

SPECIFIC EXPECTATIONS

C1. Nutrition throughout the Lifespan

By the end of this course, students will:

- C1.1** explain how growth and development throughout the lifespan (*e.g., during pre-pregnancy, pregnancy, lactation, infancy, childhood, adolescence, later life*) affect nutritional needs and food choices

Teacher prompt: “What types of food choices could fulfil the nutritional needs of young adults? Primary school children?”

- C1.2** explain how a variety of factors (*e.g., familial, social, emotional, cultural, religious, economic, geographic, ethical, psychological*) influence the food choices people make

Teacher prompts: “Do you choose different foods when eating with your peers than when eating with your family? If so, why?” “How does where you live affect your access to food?”

- C1.3** describe food and nutrition products that are designed to meet the needs of people at different stages of the lifespan (*e.g., single-serving products, baby foods, meal-replacement drinks*)

Teacher prompts: “How do food companies target people at different stages of the lifespan?” “What are some products designed to appeal to parents of young children? To athletes? To seniors?”

- C1.4** plan and prepare a food item or items appropriate to the nutritional needs of people at a specific stage of the lifespan

C2. Nutrition and Disease

By the end of this course, students will:

- C2.1** describe some eating practices that help in the prevention and management of particular health conditions (*e.g., allergies, diabetes, cardiovascular disease, osteoporosis, kidney disease*)

Teacher prompt: “What types of fat would need to be reduced or eliminated in the diet of someone with cardiovascular disease?”

- C2.2** identify social and emotional conditions that may result in unhealthy eating patterns and contribute to illness and disease (*e.g., busy schedules, expectations related to body shape, stress, scarcity of resources*)

Teacher prompt: “How does stress affect your eating habits? What might be some of the long-term consequences of regularly maintaining such a diet?”

- C2.3** explain how various factors (*e.g., heredity/genetics, socio-economic status, geography, lifestyle, activity levels*) can contribute to nutrition-related illnesses and health conditions

Teacher prompts: “Are diseases such as diabetes, heart disease, or high cholesterol always the result of heredity? What other factors might also play a role?” “How does childhood poverty affect people’s long-range health outcomes?”

- C2.4** describe and explain the reasons for the nutrient needs and dietary requirements of people with specific illnesses or diseases

(e.g., diabetes, HIV/AIDS) or people undergoing particular medical treatments (e.g., people undergoing chemotherapy, people taking particular pharmaceutical drugs or drug combinations)

Teacher prompt: “In what ways is diabetes management about much more than just sugar intake?”

- C2.5** explain the scientific basis for particular strategies to prevent food- and nutrition-related diseases and illnesses

Teacher prompt: “What are some specific strategies to prevent food- and nutrition-related diseases and illnesses that align with the recommendations in Canada’s Food Guide?”

- C2.6** plan and prepare a food item or items to meet the nutritional needs of people with a specific illness or disease

C3. Trends and Patterns in Food and Nutrition

By the end of this course, students will:

- C3.1** analyse new and emerging food- and nutrition-related products and services (e.g., additives, functional foods, whole-wheat pasta, soy products, energy drinks, vitamin-enhanced drinks, local food initiatives, agri-tourism, molecular gastronomy, the slow food movement) in terms of their real or perceived benefits to Canadian consumers (e.g., health benefits, time savings, environmental benefits)

Teacher prompts: “What new foods are being marketed now, and what is the perceived need that they are addressing?” “What are the perceived benefits of vitamin-enhanced water?” “What groups are the intended market for energy drinks?”

- C3.2** explain why people adopt various eating patterns (e.g., vegetarian diet, slow food diet, organic diet, weight-loss program)

- C3.3** describe the effects on overall health of various popular diets and food trends (e.g., low-carbohydrate diets, promotion of antioxidants and phytochemicals)

Teacher prompt: “What criteria would you use to assess whether a diet is effective and nutritionally sound?”

- C3.4** explain some ways in which scientific research on nutrition and health has influenced government legislation and policy (e.g., nutrition labelling requirements, trans-fat-reduction campaigns, school food and beverage policies, policies to implement daily physical activity in schools)

- C3.5** plan and prepare a food item or items using an ingredient that is currently being marketed as a functional food (e.g., flax seed, high-protein pasta, blueberries, pomegranates, chia)

D. LOCAL AND GLOBAL ISSUES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1. Food Security:** demonstrate an understanding of various factors involved in achieving and maintaining food security;
- D2. Food Production and Supply:** demonstrate an understanding of various factors that affect food production and supply;
- D3. Food Production and the Environment:** demonstrate an understanding of the effects of food production on the environment.

SPECIFIC EXPECTATIONS

D1. Food Security

By the end of this course, students will:

- D1.1** explain the importance of each of the key components of food security (*e.g., availability, accessibility, adequacy, acceptability, sustainability*)

Teacher prompt: “Why would access to potable water be considered a food security issue?”

- D1.2** explain how social, cultural, economic, and political factors (*e.g., gender, ethnicity, religious or political affiliation, employment, income*) contribute to nutritional inequalities among people within the same community

Teacher prompts: “In wealthy countries how is it possible that some people can go hungry?” “How is access to food sometimes used during times of conflict to subjugate groups?”

- D1.3** describe the relationships between poverty, food insecurity, poor nutrition, and poor health

Teacher prompt: “How does poverty affect one’s ability to get food?”

- D1.4** explain how various food distribution systems affect food security, locally and globally (*e.g., farmers’ markets supply local foods from identifiable sources; large supermarkets provide increased access to foods year-round but may contribute to lack of access to foods in other countries; fair-trade networks guarantee the working*

conditions of the food producers but may lead to choices to grow cash crops rather than food for local consumption)

Teacher prompt: “How do changes in demand for local foods affect the food security of farmers and communities?”

- D1.5** demonstrate the ability to act to combat food insecurity at the local and global level (*e.g., write to elected representatives or government officials; volunteer with a breakfast program; fundraise for community water wells; plant trees; buy products from women-led cooperatives; become involved in a community garden; work on a local farm*)

Teacher prompts: “What criteria will you use to determine the best course of action that you could take to fight food insecurity?” “How do women-led cooperatives help to fight food insecurity?”

D2. Food Production and Supply

By the end of this course, students will:

- D2.1** outline how geographical factors, physical conditions, and natural disasters (*e.g., climate, weather, soil conditions, proximity to water, mudslides, floods, earthquakes*) affect food supply and production

Teacher prompt: “Why are the soil conditions in the Holland Marsh so favourable to crop growth?”

D2.2 explain the effects of various agricultural methods (e.g., crop rotation, integrated pest management, fallow fields, intercropping, no tillage) on local or global food production and yields

Teacher prompts: “Why might regular tillage of soil decrease crop yields?” “How can leaving a field fallow for a season lead to increases in crop yields in future years?”

D2.3 explain the effect of various economic, social, and political factors (e.g., debt-repayment obligations, demand for cash crops, oil prices, free-trade agreements, trade embargos or bans, controls on fishing and hunting, import-export restrictions designed to prevent or control outbreaks of disease) on food supply and production

Teacher prompt: “How do fluctuations in the price of oil on world markets affect food production?”

D2.4 analyse the effect of various trends in agriculture and aquaculture (e.g., organic farming, use of antibiotics, fish farming, genetic engineering, greenhouse food production) on local and global food supply and production

Teacher prompts: “How has genetic engineering affected the production and consumption of food?” “What regulations have Health Canada and the Canadian Food Inspection Agency put in place regarding the use of antibiotics with Canadian livestock? How are the regulations different for organic farming in Canada? How might the differences between organic and traditional farming practices with respect to the use of antibiotics lead to differences in the amount and quality of meat produced?”

D3. Food Production and the Environment

By the end of this course, students will:

D3.1 describe how consumer food choices affect the environment, locally and globally (e.g., demand for imported food increases the

amount of energy used in transportation; choice of overpackaged products increases the volume of waste going to landfills; choice of fair-trade products supports sustainable farming and small-scale farmers; demand for local produce supports farmers’ markets and reduces use of fossil fuels)

Teacher prompt: “What is the environmental impact of purchasing overpackaged foods? Of purchasing bottled water?”

D3.2 explain the effect on the environment of various agricultural trends (e.g., growing crops for biofuels) and food-production technologies (e.g., types of farm equipment, types of energy sources, climate-control techniques, genetic engineering of foods)

Teacher prompt: “What are some positive and negative environmental effects associated with the production and consumption of genetically modified foods?”

D3.3 explain the effect of various environmental protection laws and regulations on food supply and production (e.g., policies related to forest preservation, fuel emission standards, pesticide use)

Teacher prompts: “What impact have cod-fishing bans on the east coast of Canada had on the fishing and fish-processing industries?” “How do current fishing and hunting bans affect relationships between Aboriginal and non-Aboriginal people in Canada?”

D3.4 demonstrate an understanding of health, safety, and environmental issues related to food supply and production (e.g., risks associated with bioaccumulation of pesticides and hormones, risks of contamination during food production), and identify legislation that is designed to protect Canadian consumers (e.g., Canada Agricultural Products Act, Food and Drugs Act)

Teacher prompts: “How can consumer awareness of the food-production process benefit food producers, consumers, and the environment?” “What are the health risks associated with the use of bisphenol A?”

E. FOOD-PREPARATION SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1. Kitchen Safety:** demonstrate an understanding of practices that ensure or enhance kitchen safety;
- E2. Food Safety:** demonstrate an understanding of practices that ensure or enhance food safety;
- E3. Food Preparation:** demonstrate skills needed in food preparation.

SPECIFIC EXPECTATIONS

E1. Kitchen Safety

By the end of this course, students will:

- E1.1** describe common accidents that can occur in the kitchen (*e.g., cuts, burns, fires, falls, poisoning, electric shocks*)
- E1.2** demonstrate an understanding of safe practices within the food-preparation area (*e.g., safely handle hot foods; prevent spatters, scalds, and cuts; wipe up spills immediately*)
- E1.3** demonstrate an understanding of appropriate emergency responses to common accidents associated with food preparation (*e.g., cuts, burns, scalds, fires*)

E2. Food Safety

By the end of this course, students will:

- E2.1** outline the causes and symptoms of food-borne illnesses (*e.g., E. coli poisoning, botulism poisoning, Clostridium perfringens poisoning, salmonellosis, listeriosis*) and techniques for preventing these illnesses
- E2.2** use appropriate personal hygiene practices to prevent contamination of food (*e.g., wash hands frequently; cover a cough or sneeze in their sleeve; use gloves to cover cuts or wounds; tie hair back*)
- E2.3** use safe food-handling practices to prevent cross-contamination by pathogens, parasites, and allergens in the food-preparation area (*e.g., wash fresh produce; sanitize cutting boards after contact with meat products; sanitize implements that come into contact with allergens when preparing food for or with people with known allergies; sanitize work surfaces; replace or sanitize sponges or cloths frequently; use proper clean-up procedures*)

- E2.4** follow appropriate protocols to ensure food safety (*e.g., cook foods to recommended temperatures; keep hot foods hot and cold foods cold; store food appropriately; wipe tops of cans before opening; check "best-before" dates; demonstrate awareness of common allergenic ingredients*)

E3. Food Preparation

By the end of this course, students will:

- E3.1** identify and select appropriate tools, equipment, and ingredients for use in food preparation
- E3.2** demonstrate the ability to follow a recipe
- E3.3** demonstrate the ability to adapt recipes to accommodate specific dietary needs
- E3.4** demonstrate the ability to safely use, maintain, clean, and store tools and equipment used in food preparation
- E3.5** demonstrate the ability to measure quantities accurately (*e.g., use different strategies for measuring wet and dry ingredients; level off excess amounts; measure liquids at eye level*)
- E3.6** demonstrate the correct use of food-preparation techniques (*e.g., stirring, beating, whipping, chopping, broiling, frying*)
- E3.7** demonstrate the ability to manage time effectively in food preparation
- E3.8** plan, prepare, and serve a food item or items according to set criteria